

- 7 The Sumatran tiger, *Panthera tigris sumatrae*, is classified as critically endangered on the International Union for Conservation of Nature (IUCN) Red List of Threatened Species™.

Fig. 7.1 shows a Sumatran tiger.



Fig. 7.1

(a) Fig. 7.2 shows the number of wild Sumatran tigers in the world between 1970 and 2020.

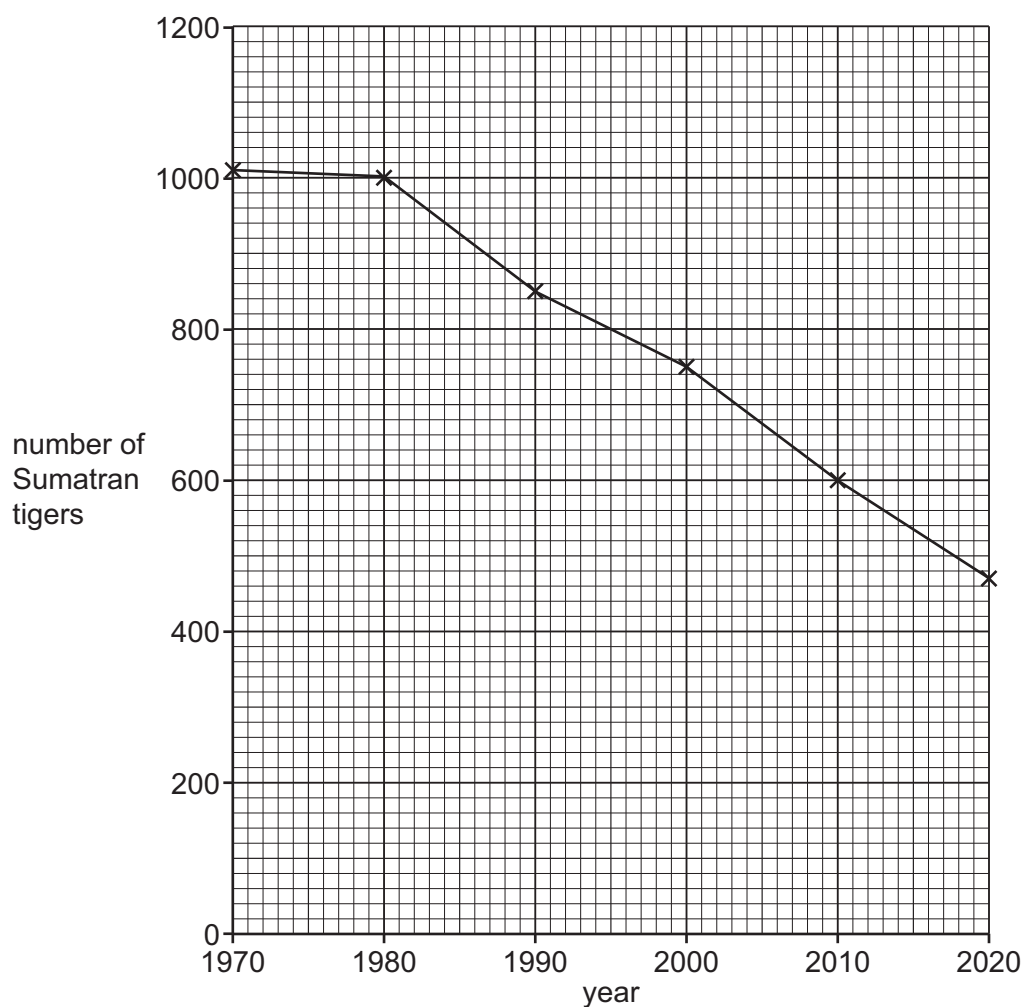


Fig. 7.2



- (i) Calculate the mean rate of decrease in the Sumatran tiger population between 1970 and 2020.

mean rate of decrease = year⁻¹ [2]

- (ii) There are a number of different ways to help conserve Sumatran tigers. For example, some zoos have captive breeding programmes.

Outline ways in which Sumatran tigers may be conserved, **other** than captive breeding programmes.

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..... [3]

- (b) Outline reasons for maintaining animal biodiversity.

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..... [4]

(c) Captive breeding programmes sometimes use IVF.

Table 7.1 shows some of the events that occur during an IVF procedure.

They are **not** listed in the correct order.

Table 7.1

letter	event
A	sperm added to oocyte
B	embryo formed
C	female given hormones to stimulate ovulation
D	zygote placed in culture medium
E	embryo placed in uterus of female
F	zygote formed
G	oocyte harvested with a fine needle

Complete Table 7.2 to show the correct order of the events.

One of the events has already been added in the correct position.

Table 7.2

correct order	letter
1	
2	
3	
4	F
5	
6	
7	

[4]

[Total: 13]



Question 8 starts on page 26.